

April 10, 2022

Dr. Stephanie Johnson
U.S. Department of Energy
Office of Energy Efficiency and Renewable Energy
Building Technologies Office, EE-5B
1000 Independence Avenue SW
Washington, DC 20585-0121



NEEA Comments on the Request for Information regarding Test Procedures and Energy Conservation Standards for Consumer Air Cleaners

Docket Number: EERE-2021-BT-STD-0035 and EERE-2021-BT-TP-0036

Dear Dr. Johnson:

The following comments are submitted on behalf of the Northwest Energy Efficiency Alliance (NEEA). NEEA is a non-profit organization representing an alliance of more than 140 Northwest utilities and energy efficiency organizations working on behalf of more than 13 million energy consumers.

NEEA has been promoting the market adoption of energy efficient consumer air cleaners in the Northwest since 2015 as part of the U.S. Environmental Protection Agency (EPA) ENERGY STAR® Retail Products Platform (ESRPP). NEEA strongly supports the U.S. Department of Energy (DOE)'s efforts to adopt test procedures and energy conservation standards for this product category and are pleased share RPP sales data and program insights to support DOE's analysis.

Thank you for the opportunity to provide a response to DOE on the Request for Information (RFI) for consumer air cleaners. NEEA has compiled market insights based on implementation of energy efficiency programs for consumer or room air cleaners to address DOE's questions about product definitions, classifications, sales, efficiency performance, incremental costs, energy savings, operating hours, and lifetime. We respectfully submit these comments for your consideration.

Background on NEEA's ENERGY STAR Retail Product Platform Data

NEEA's ESRPP program is an innovative concept in energy efficiency program design. Coordinating with EPA ENERGY STAR, the platform works directly with corporate-level national retailers to provide mid-stream incentives on qualified energy-efficient products. These incentives influence retail buying and stocking practices, to ultimately drive manufacturing and higher product standards and specifications for a portfolio of energy-efficient products sold through the retail channel.

In exchange for the incentives, retailers provide full-category sales data for all products in the portfolio. NEEA uses this sales data to conduct extensive analysis of the efficiency profile of each product category, as well as identifying market trends and efficiency opportunities. Through ESRPP, NEEA works with a variety of national retailers, which collectively have over 300 store locations in NEEA territory. The NEEA air cleaner data shared in these comments includes sales from Target, Best Buy, Costco, Home Depot, KMart, Lowe's, and Sears in the states of Idaho, Montana, Nevada, Oregon, and Washington.

NEEA routinely analyzes the retail sales data and online product information to understand the air cleaners market and how it is evolving over time. NEEA provided incentives for air cleaners from 2015 through the first quarter of 2019 and so the data insights shared below reflect market conditions for this period. NEEA has bolstered this incentive program data with recently compiled statistics from other data sources.

Compiled Market Insights

To support DOE's Market and Technology Assessment, NEEA has compiled data from RPP as well as other credible sources. The significant number of air cleaners sold historically, and their associated energy use combined with sales increases in response to wildfires and the pandemic indicate this a compelling product category for DOE regulation. Table 1 below shows the quarterly NEEA RPP sales of room air cleaners by year.

Table 1: NEEA Quarterly ESRPP Sales by Year

Year	Q1 Sales	Q2 Sales	Q3 Sales	Q4 Sales
2015	7,920	4,993	4,438	4,155
2016	7,156	4,309	4,390	7,313
2017	9,548	5,958	10,920	10,031
2018	8,841	7,460	11,382	12,381
2019	3,789			

Given the significant market significant shifts since 2019, NEEA has compiled information from other sources to support DOE's research efforts.

The AHAM directory of certified room air cleaners (<https://www.ahamdir.com/room-air-cleaners/>) captures a total of 84 brands and 768 models, 517 of which are ENERGY STAR certified.

In 2018, [EPA shipment data](#) indicated there were 5 million shipments nationally, with 1.7 million of those shipments being ENERGY STAR certified. The most recent ENERGY STAR unit shipment data¹ published indicates that 6.5 million ENERGY STAR certified room air cleaners were shipped in 2020, though EPA was unable to estimate the total shipments or ENERGY STAR market penetration for air cleaners in 2020 due to dramatic changes in the market that are inconsistent with previous market trends. This report includes data from 87 percent of the 60 ENERGY STAR partners for this product category. The 2020 ENERGY STAR certified shipments reflect models that meet the Version 2.0 ENERGY STAR specification, which became effective on October 17, 2020. The 2018 ENERGY STAR certified shipments reflect models that met the less-stringent Version 1.2 ENERGY STAR specification.

Some inconsistencies in market trends are likely attributed to increased sales in response to the COVID-19 pandemic and wildfires. On April 22, 2021, AHAM published a blog post² that shared some insights from a recent consumer survey on appliance use during COVID. Those who responded to AHAM's survey reported using their

¹ ENERGY STAR® Unit Shipment and Market Penetration Report Calendar Year 2020 Summary, https://www.energystar.gov/sites/default/files/asset/document/2020%20USD%20Summary%20Report_Lighting%20%20EVSE%20Update.pdf

² AHAM Blog Post, "The Great Indoors: What we've learned about air cleaners during COVID-19", <https://blog.aham.org/the-great-indoors-what-weve-learned-about-air-cleaners-during-covid-19/>

room air cleaners 42 percent more than they did before the pandemic. Sixteen percent reported having purchased an air cleaner during the pandemic.

A Bloomberg article from August 2021 states that the market witnessed a growth rate of 57 percent in 2020 in response to COVID-19 and that US sales continue to climb due to wildfires on the West Coast. Coway purifiers is quoted as seeing their sales increase by 102.8 percent in May, June, and July compared to the same period in 2020.³

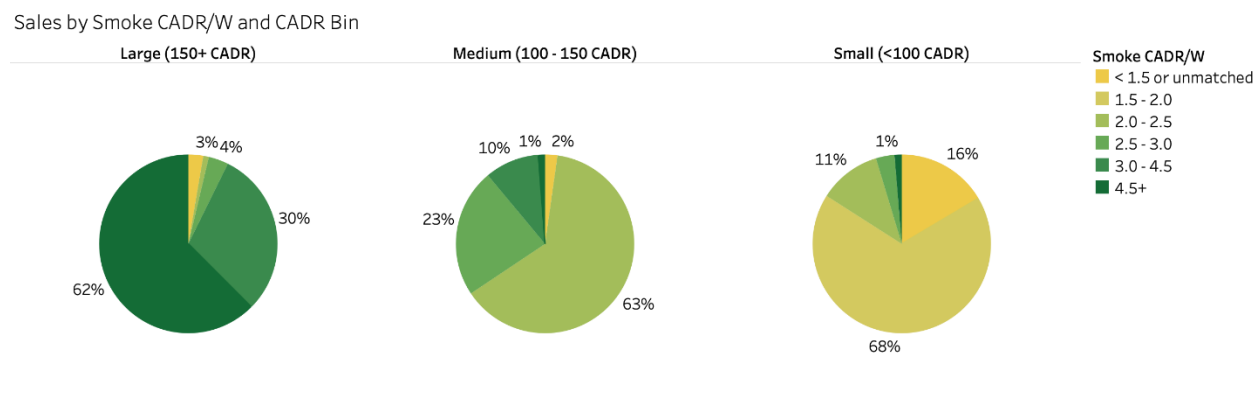
NEEA Supports DOE's Proposed Scope and Definition (Section II.A.)

In the interest of maximizing energy savings to consumers, NEEA supports DOE's proposal to use a broad definition for room air cleaners that includes both portable and mounted units. NEEA also supports excluding other products that fit into alternate DOE product categories, namely those that are classified as "lamps primarily designed to produce radiation in the ultraviolet region of the spectrum."

NEEA Recommends Product Class Distinctions Based on CADR and Smoke CADR/W as the Efficiency Metric (Section III.A.1)

Review of NEEA RPP sales data for room air cleaners demonstrates a notable trend in efficiency by clean air delivery rate (CADR). Looking at sales from the ESRPP data, there is a statistically significant relationship between smoke CADR bin and CADR/W, with a CADR increase of 100 on average corresponding to a CADR/W increase of 1.17. Looking at this correlation, the p-value is < 0.0001, which means that within the northwest market, there was a strong relationship between size and efficiency between 2015-2019.

Chart 1: Sales by Smoke CADR/W and CADR Bin



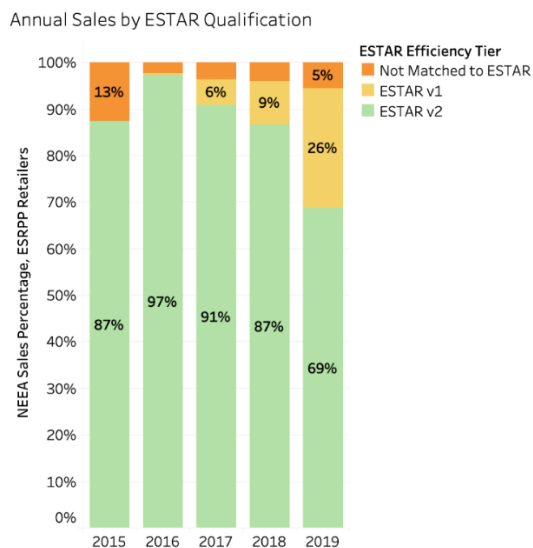
These categories are consistent with the ENERGY STAR specification, which has established separate performance requirements for three Smoke CADR Bins: $30 \leq \text{CADR} < 100$, $100 \leq \text{CADR} < 150$, and $\text{CADR} \geq 150$. NEEA recommends that DOE establish product class distinction by CADR.

³ Bloomberg Article, "Covid-19 and Wildfires Spell Big Business for the Air Purifier Industry", August 5, 2021. <https://www.bloomberg.com/news/articles/2021-08-05/covid-19-and-wildfires-are-driving-a-big-increase-in-u-s-air-purifier-sales>

NEEA Recommends Using ENERGY STAR Version 2.0 Specification Requirements as Baseline Efficiency Level (Section III.C.1)

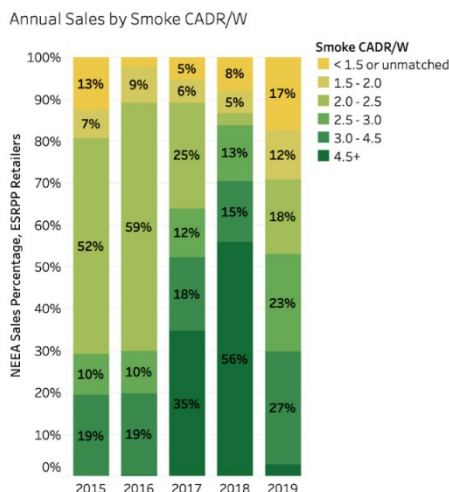
Should DOE pursue the efficiency-level approach to analysis, NEEA strongly recommends that the DOE use the ENERGY STAR Version 2.0 specification as the baseline efficiency level. The high percentage of sales of ENERGY STAR Version 2.0 models seven years ago is noteworthy. Chart 2 indicates that 87 percent of the 2015 room air cleaner sales sold through ESRPP-participating retailers in the northwest met the ENERGY STAR Version 2.0 requirements. The fluctuations in percent of room air cleaner sales with performance equal to or greater than ENERGY STAR Version 2.0 requirements are likely due to changes in incentive program offerings and retailer participation. Even at their lowest, ENERGY STAR Version 2.0 units represent well over half of sales in NEEA's territory, demonstrating the market's ability to provide product at this level of efficiency. This data provides a compelling basis to serve as baseline efficiency.

Chart 2: Annual Sales by ENERGY STAR Qualification



To support DOE's efforts to identify other performance clusters, Chart 3 provides insights into the specific Smoke CADR/W of the room air cleaners sold.

Chart 3: Annual Sales by Smoke CADR/W



NEEA Shares Incremental Cost Data to Support DOE's Analysis (Section III.C.2)

NEEA developed a memo on Incremental Cost of ENERGY STAR Version 2 Air Purifiers, Matched Pair Analysis, which found incremental cost to be small. Large capacity room air cleaners were found to have an incremental price of \$6.00, while smaller capacity units typically saw increases of closer to \$26.00. The incremental cost memo has been attached to these comments for DOE's consideration.

NEEA Offers Energy Savings Estimates, Hours of Use, and Lifetime Assumptions (Section III.E.)

The energy savings estimates calculated from unit energy savings (UES) and shipments range from 1.5 – 13 TWh/year of savings, which represents all air cleaners in stock in the U.S. Assuming a product lifetime of 9 years and 30 years of shipments, this is equivalent to 0.15 – 1.3 quads saved.

In calculating UES, [EPA](#) indicates that savings could be as much as 120 kWh/year if consumers operate their air cleaners continuously throughout the year. NEEA has typically estimated average operation to be 8 hours per day, which is based on seasonal operation or part-day operation and results in savings closer to 40 kWh/yr.

Another source for hours of operation is the Northwest Regional Technical Forum (RTF). [The RTF](#) estimates 5,844 hours per year (16 hours/day) for room air cleaners. The RTF's measure workbook references the IMC analysis, cites a lifetime of 9 years, and estimates UES to be 150 kWh/year.

Based on review of the most efficient models found on the ENERGY STAR Product Finder for air cleaners, NEEA has developed recommendations for max-tech level in Table 2.

Table 2: Most Efficient Models on the ENERGY STAR Product Finder for Air Cleaners

Model Name	Capacity Bin (CADR)	Efficiency (Smoke CADR/W)
Powermatic - PM1250 and JET - JAFS-1250	> 150	14.8
Blueair - Joy S	< 150 > 100	12.4
Envion,Ionic Elite - TA750	< 100	10.3

We hope our comments and data provide desired market and technology insight to support DOE's assessment of standards for consumer air cleaners. NEEA welcomes the opportunity to discuss our comments further and looks forward to continued engagement throughout the DOE standards development process.

Sincerely,

A handwritten signature in black ink, appearing to read "Nicole Dunbar".

Nicole Dunbar, P.E.

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